

Experiment 1- 6 outcomes

Experiment 1:

```
PS C:\Users\DELL\country\Scripts\city> cd C:\Users\DELL\OneDrive\Documents
PS C:\Users\DELL\OneDrive\Documents> python "import nltk.py"
[nltk_data] Error loading punkt: <urlopen error [WinError 10065] A
[nltk_data]      socket operation was attempted to an unreachable host>
[nltk_data] Error loading punkt_tab: <urlopen error [Errno 11001]
[nltk_data]      getaddrinfo failed>
[nltk_data] Error loading wordnet: <urlopen error [Errno 11001]
[nltk_data]      getaddrinfo failed>
Enter a sentence: The boys were running happily.
```

Original Text:

The boys were running happily.

Tokens:

['The', 'boys', 'were', 'running', 'happily', '.']

Stemmed Words:

['the', 'boy', 'were', 'run', 'happili', '.']

Lemmatized Words:

['The', 'boy', 'were', 'running', 'happily', '.']

Comparison:

Stemming reduces words to root forms, which may not be meaningful.

Lemmatization converts words to meaningful base forms.

Experiment 2:

```
Enter a sentence: Natural Language Processing improves text analysis
```

Tokens:

['Natural', 'Language', 'Processing', 'improves', 'text', 'analysis']

POS Tags:

Natural -> JJ

Language -> NNP

Processing -> NNP

improves -> VBZ

text -> JJ

analysis -> NN

Tag Meanings:

NN -> Noun

VB -> Verb

JJ -> Adjective

RB -> Adverb

PRP -> Pronoun

DT -> Determiner

Total Words: 6

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Experiment 3:

```
PS C:\Users\DELL\OneDrive\Documents> python exp3.py
[nltk_data] Downloading package wordnet to
[nltk_data] C:\Users\DELL\AppData\Roaming\nltk_data...
[nltk_data] Package wordnet is already up-to-date!
[nltk_data] Downloading package omw-1.4 to
[nltk_data] C:\Users\DELL\AppData\Roaming\nltk_data...
Enter number of headlines: 4
Enter headline: Stock market gains record profits
Enter headline: Investors celebrate rise in share prices
Enter headline: Football team wins championship
Enter headline: Cricket tournament attracts audience

Cosine Similarity Matrix:
[[1. 0. 0. 0.]
 [0. 1. 0. 0.]
 [0. 0. 1. 0.]
 [0. 0. 0. 1.]]

Headline Clusters:
Stock market gains record profits -> Cluster 0
Investors celebrate rise in share prices -> Cluster 0
Football team wins championship -> Cluster 0
Cricket tournament attracts audience -> Cluster 1

Enter first word: cricket
Enter second word: wins
WordNet Similarity: 0.05263157894736842
```

Experiment 4:

```
PS C:\Users\DELL\OneDrive\Documents> python "exp4.py"
Enter number of documents: 3
Enter document: Machine learning improves healthcare diagnosis
Enter document: Deep learning is used in artificial intelligence
Enter document: Climate change affects biodiversity

Enter search query: artificial intelligence

TF-IDF Similarity Scores:
Document 1 : 0.0
Document 2 : 0.551
Document 3 : 0.0

LSA Similarity Scores:
Document 1 : 1.0
Document 2 : 1.0
Document 3 : 0.0

Most Relevant Document:
Machine learning improves healthcare diagnosis
```

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Experiment 5:

```
PS C:\Users\DELL\OneDrive\Documents> python "exp5.py"
[nltk_data] Downloading package punkt to
[nltk_data]   C:\Users\DELL\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to
[nltk_data]   C:\Users\DELL\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt_tab is already up-to-date!
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data]   C:\Users\DELL\AppData\Roaming\nltk_data...
[nltk_data]   Package averaged_perceptron_tagger_eng is already up-to-
[nltk_data]   date!
Enter legal text: John Smith filed a case against ABC Corporation in New York Court

Detected Named Entities:
John -> ENTITY
Smith -> ENTITY
ABC -> ENTITY
Corporation -> ENTITY
New -> ENTITY
York -> ENTITY
Court -> ENTITY

Enter actual number of entities: 8

Predicted Entities: 7
NER Accuracy: 87.5 %
```

Experiment 6:

```
PS C:\Users\DELL\OneDrive\Documents> python "exp6.py"
[nltk_data] Downloading package punkt to
[nltk_data]   C:\Users\DELL\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to
[nltk_data]   C:\Users\DELL\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt_tab is already up-to-date!
Enter biomedical sentence: Aspirin reduces fever in patients
Actual Relation (1/0): 1

Tokens:
['aspirin', 'reduces', 'fever', 'in', 'patients']

Predicted Relation: 1
Precision: 1.0
Recall: 1.0
F1-Score: 1.0
```